Central Limit Theorem Formula

❖ Let us assume we have a random variable X. Let σ be its standard deviation and μ is the mean of the random variable.

❖ Now as per the Central Limit Theorem, the sample mean $\bar{X}$ will approximate to the normal distribution which is given as $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$.

❖ The Z-Score of the random variable $\bar{X}$ is given as Z . Here $\bar{x}$ is the mean of $\bar{X}$. The image of the formula is attached below.

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Sample Mean = Population Mean = μ

$$\text{Sample Standard Deviation} = \frac{\text{Standard Deviation}}{n}$$

OR

$$\text{Sample Standard Deviation} = \frac{\sigma}{\sqrt{n}}$$

Applications of the Central Limit Theorem (CLT)

- ❖ **Hypothesis Testing** – Ensures sample means follow a normal distribution for statistical tests.
- ❖ **Confidence Intervals** – Helps estimate population parameters with a range of values.
- ❖ **Quality Control** – Monitors manufacturing consistency using sample means.
- ❖ **Election Polling** – Predicts voting outcomes from small survey samples.
- ❖ **Financial Market Analysis** – Models stock returns and investment risks.
- ❖ **Machine Learning** – Used in resampling, bootstrapping, and feature scaling.
- ❖ **Insurance & Risk Management** – Predicts claim amounts and sets premiums.
- ❖ **Medical Research** – Generalizes clinical trial results for larger populations.
- ❖ **Supply Chain Management** – Forecasts demand and optimizes inventory.
- ❖ **Fraud Detection** – Identifies anomalies in financial transactions